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Flow duration of a dike constrained by palaeomagnetic data

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SUMMARY

We have conducted a palaeomagnetic study of a Permian red sandstone layer intruded by a 70 cm wide Plio-Pleistocene basaltic dike, with the ultimate goal of estimating the flow duration of the magma before its solidification.

31 samples were collected at various distances from the dike/sandstone interface and were thermally demagnetized in order to separate the overprinted Plio-Pleistocene magnetization component from the primary Permian one and to determine for each sample the unblocking temperature, T_{ub}, necessary to erase totally the overprint. This temperature can be used to determine the maximum palaeotemperature reached at any distance from the contact, provided the secondary magnetization is a (P)TRM and the thermal fluctuation effects and actual cooling rates are taken into account.

Examination of various magnetic parameters (NRM intensity, 15 days VRM intensity, initial magnetic susceptibility and hysteresis curves) allowed us to establish the nature of the overprinted magnetization and consequently to exclude samples carrying a CRM overprint. An apparent palaeotemperature profile and two corrected ones were thus deduced from the samples in which the dike emplacement produced no crystallization of new magnetic minerals.

These profiles were compared with the calculations of a new conductive thermal model which includes the basalt's latent heat of crystallization, the flow duration of the dike, the width of the circulating magma, and the temperature dependence of the thermal conductivity, k, in the basalt and in the sandstone, or alternatively convection simulated by an increase of the k values.

Good fits with the experimental data are obtained for a flowing width consistent with field observations and when the latent heat is taken into account, in the two following cases. First, when k is temperature dependent in the basalt and in the sandstone, which implies a magma flow duration of 5 days. Second, when a limited convection process is assumed in the sandstone only; here, a flowing time of only 1 day is required.

The corrected palaeotemperature profiles argue in favour of the second hypothesis, suggesting that when a dike is emplaced, heat transfer by fluids may play an important role in the surrounding rock.

Key words: blocking/unblocking temperature, conduction, convection, CRM, palaeotemperature, thermal modelling, TRM.

1 INTRODUCTION

The intrusion of magmatic material into sedimentary rocks produces a thermal metamorphism and a partial to total remagnetization. The metamorphic transformations should theoretically provide a profile of the maximum temperatures reached in the host rock as a function of the distance from the dike. What is required is a minimum number of key metamorphic reactions and knowledge of the fluid and gas pressures. In nature, these conditions are rarely fully satisfied. Four example, Fabre (1988) and Fabre, Kast & Girod (1989) propose a thermal model of cooling based on the negative correlation between the thickness of the chilled margin of a dike and the magma flow duration. However, they could not observe enough metamorphic reactions to test fully their cooling model.

Palaeomagnetism can be a more powerful method than petrology to decode the palaeotemperatures recorded by a rock, provided the remagnetization of the rock is a partial or a total thermoremanent magnetization (TRM). If so, the usual stepwise demagnetization technique used in palaeomagnetism provides the unblocking temperature (T_{ub}) spectra of the different magnetization components carried by the rock samples. The maximum unblocking temperature ($T_{ub\,max}$) of the overprinted magnetization can be related to the maximum temperature suffered by the host rock subsequently to the thermal event. Consequently, palaeotemperatures reached in the surrounding rock can be determined at each sampling point away from the dike border. The palaeo-T profile so obtained can be compared to those calculated by thermal modelling.

Others studies have already attempted to compare thermal models of cooling with palaeotemperature profiles determined from palaeomagnetic studies. These studies where concerned with the heating of a basaltic basement by an overlying cooling lava (Audunsson & Levi 1988), or the effects on a sandstone of an intrusive body (Robinson & McClelland Brown 1987; Kipfer & Heller 1988), or the influence on lava flows of intruded dykes (Buchan *et al.* 1980; McClelland Brown 1981). However, none of these studies attempted to assess the nature of the overprinted magnetization.

In this paper, we will first try to determine the palaeotemperature profile in the host rock near a basaltic dike. The necessary preliminary step will be to ascertain that the overprinted magnetization is indeed a TRM. Then the palaeotemperature profile will be compared to those calculated from a thermal conductive model that allows us to examine the effects of various parameters and to estimate the flow duration in the fissure before the solidification of the magma.

2 GEOLOGICAL ASPECTS, SAMPLING AND EXPERIMENTAL PROCEDURES

2.1 Geological setting

The subject of this work is a Saxonian red sandstone outcropping in the Permian 'Bassin de Lodève' southwest of the French Massif Central (site coordinates: 43.7°N; 3.4°E). It is one of the sandstone layers intercalated in the thick homogeneous pile of finer red silty sandstones. The whole sequence was intruded by a basaltic dike during a Plio-Pleistocene volcanic episode.

The outcrop is fresh and well exposed in a road cut. The sandstone layer is 30 to 40 cm thick, fine grained, reddish to purple and slightly south-dipping. It looks homogeneous except for the 3 to 5 cm which are in contact with the dike. Here, the yellowish colour and spongy texture of the rock indicates a strong alteration. The dike is 70 cm wide, subvertical and slightly sinuous up to the erosion level of the outcrop. Its central part (45 to 50 cm wide) is made of fine-grained, dark and fresh-looking, massive and radially fissured basalt. The borders (10 to 13 cm on each side) show differenciated concentric layers of altered material. There is a total lack of any glassy margin. It is unlikely that there was once glass here that has altered with time. In other places in the Bassin de Lodève, Fabre (1988) has described basaltic dikes of the same age exhibiting both well-preserved chilled glassy margins and no glass at all.

2.2 Sampling description

Five basalt samples of 1 inch in diameter were drilled from the central fresh part of the dike in order to give the Plio-Pleistocene palaeomagnetic reference direction. In the sandstone, 31 samples were cored from the same side of the dike. Their distances from the dike/sediment interface were carefully measured and were corrected for the skewness of the road cut which is not perpendicular to the dike direction. The sampling extends to over 8 m but is mainly concentrated and regularly spaced between 1.3 and 120 cm. Two or three specimens of standard height 2.2 cm were drawn from each sample.

2.3 Experimental procedure

Prior to any demagnetization, we performed a magnetic viscosity test (Thellier & Thellier 1959). This test (Prévot 1981), determines the intensity of a viscous remanent magnetization (VRM) acquired by a specimen in 15 days in the earth's magnetic field and compares it with the 'stable NRM', that is the original natural remanent magnetization from which the 15 days VRM is substracted. The viscosity coefficient, %v, is: VRM (15 days)/stable NRM, expressed as a percentage.

The NRM was measured with a CTF cryogenic magnetometer after the samples have been stored for 15 more days in zero magnetic field.

In order to determine accurately the metamorphic temperatures attained by the sandstone, we used a detailed thermal demagnetization procedure with narrow temperature steps. The procedure involved 21 heatings from 80° to 700 °C, with steps of 30° to 40 °C. The samples were kept about half an hour at the maximum temperature. A Pyrox non-magnetic non-inducing electrical furnace was used for the heating; the residual field inside the cooling chamber is less than about 20 nT.

Orthogonal projection diagrams were used to visualize and to identify the components of magnetization. Their directions were calculated by the least-squares method of Kirschvink (1980).

The magnetic susceptibility of the sandstone specimens was measured at room temperature prior to and after heating, using a Bartington MS2 magnetic susceptibility meter operating a 460 Hz with a 80 A m^{-1} alternating magnetic field (0.1 mT induction).

Hysteresis cycles were determined on sandstone specimens of 2.9 cm^3 volume, up to a 1.5 T maximum induction. Induced and remanent magnetizations together with induced and remanent coercive forces were measured on a translation inductometer situated between the two poles of an electromagnet.

3 IDENTIFICATION OF THE DIFFERENT MAGNETIZATION COMPONENTS

3.1 NRM

The NRM directions (Fig. 1a) of both the dike and the sandstone are reversely magnetized making identification of

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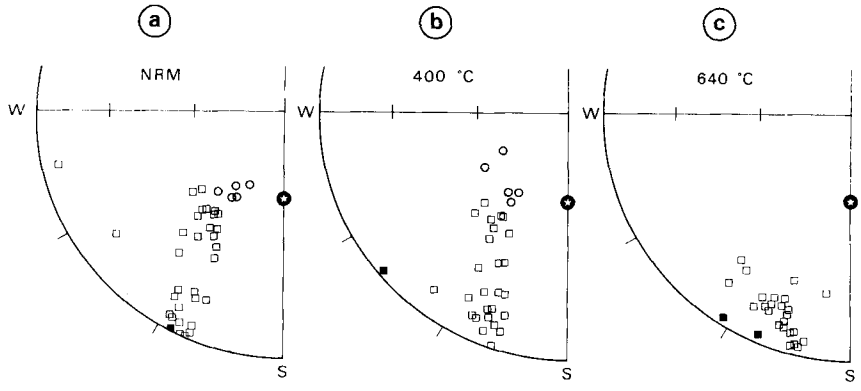


Figure 1. Equal area projection of the magnetization directions at three different demagnetization steps in geographic coordinates (before tilt correction). Circles: basalt samples; squares: sandstone samples; stars: reversed axial dipole field direction. Full symbols: positive inclination; open symbols: negative inclination.

a possible viscous magnetization, acquired *in situ,* quite simple. The directions are roughly spread along a great circle over a 60° arc from a steep inclination slightly west of the dipole field direction, in the dike and in the sandstone samples closest to the dike, to a nearly horizontal south–southwest direction characteristic of the European Permian reversed direction (Westphal *et al.* 1986), in the most remote sandstone samples.

3.2 Thermal demagnetization

The magnetic viscosity of the basalt samples in negligible ($v \leq 1.5$ per cent) and there is nearly perfect stability of the magnetization direction throughout the thermal treatment up to 400° to 500 °C, depending on the sample, beyond which no magnetization remains. The shape of the demagnetization curves suggests the presence of Ti-poor titanomagnetites. The NRM can be considered to be a pure TRM. The Plio-Pleistocene direction of the dike is $D = 208.7°$, $I = 57.3°$ (Table 1), which is consistent with the radiometric ages of 0.7 to 2 Myr obtained by Gastaud (1981) on other dikes and flows from the 'Bassin de Lodève'. Reversed magnetization directions also belonging to the Matuyama period have been reported by Ildefonse *et al.* (1972) in other volcanic units from the same area. The 15° departure from the reverse dipole field direction can be explained by secular variation.

As expected, the sandstone samples display a gradual transition of the magnetization directions (Fig. 1), from a steeply inclined direction close to the dike, to a shallow nearly Saxonian direction farther from the dike. The direction changes are visible mainly on the inclination [vertical plane of the orthogonal projection diagrams, Figs 2(b), 3(b) and 4(b)].

Sample 6, adjacent to the dike, is totally remagnetized and displays only one Plio-Pleistocene magnetization component with a narrow blocking temperature spectrum characteristic of magnetite. All of the other samples have large T_{ub} spectra irregularly spread, mainly from about 150 °C to about 670 °C (Figs 2b, c, 3b, c and 4b, c). The changes of magnetization with heating suggest the presence of two magnetic phases (Figs 2c, 3c and 4c). The twofold decrease in the demagnetization curves is not an artefact related to the change from the Plio-Pleistocene to the Saxonian direction. In the first place, it always occurs at about the same temperature while the reheating involves lower and lower T_{ub} away from the dike. Secondly, the twofold decrease persists on the demagnetization curves when the intensities of the difference vectors between consecutive heating steps are plotted against temperature.

The orthogonal projection diagrams show that in the samples closest to the dike, the transition from the Plio-Pleistocene to the Saxonian direction is rather sharp, indicating that there is almost no overlap of the unblocking temperatures of these components (Fig. 2b). For these samples, the Saxonian direction is poorly defined owing to the very few heating steps and the small amount of remanent magnetization left. Beyond about $x = 25$ cm from the dike border, the transition between the Plio-Pleistocene and Saxonian directions is of two types: in a few samples it is progressive, which is characteristic of a true T_{ub} overlap; in the others, it is chaotic with a dispersion of the vector

Table 1. Mean direction of the magnetization components in the dike and in the sandstone.

Component	N	D	I	α_{95}	K	R	Tilt Correction
Plio-Pleistocene (dike)	5	208.7	-57.3	3.6	456	4.99	No
Plio-Pleistocene (sandstone)	22	217.7	-48.0	2.2	194	21.9	No
Saxonian (sandstone)	20	202.8	-11.2	3.8	74	19.7	No
Saxonian (sandstone)	20	203.9	-20.4	3.8	73	19.7	Yes

N=number of samples ; D=Declination ; I=Inclination ; the Fisher's parameters (Fisher, 1953) are : α95=95% confidence angle ; K=precision parameter ; R=resultant vector.

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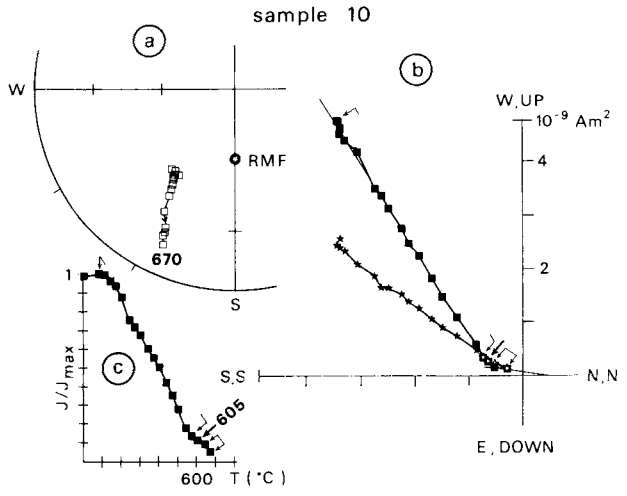


Figure 2. Thermal demagnetization of sandstone sample 10(x = 10.6 cm from the dike border) in geographic coordinates. (a) Equal area projection; RMF = reversed axial dipole field; (b) orthogonal projection diagram; squares: vertical plane; stars: horizontal plane; the solid lines represented on the vertical plane correspond to the pure Plio-Pleistocene and Saxonian magnetization components determined between the thin arrows; the thick arrow (palaeo-T_{app}) represents the middle of the temperature interval of the transition zone between both components; (c) normalized demagnetization curve.

directions. In this latter case, it is likely that a spurious magnetization has been acquired during the thermal treatment in the laboratory. This is particularly obvious for the most remote samples (28 to 33) where spurious magnetization directions are recorded systematically between 100° and 300 °C.

All the samples from $x = 4$ to 120 cm (samples 7 to 33) exhibit two or three components of magnetization (Figs 2b, 3b and 4b) as follows.

(i) A minor VRM carried by the grains having the lowest T_{ub}, easily erased by heating to a maximum temperature of 140 °C, and often less.

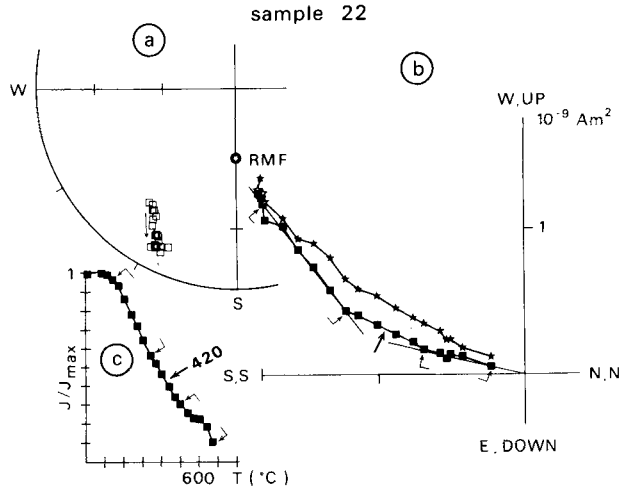


Figure 3. Same legend as for Fig. 2, for sample 22 (x = 54.2 cm from the dike border).

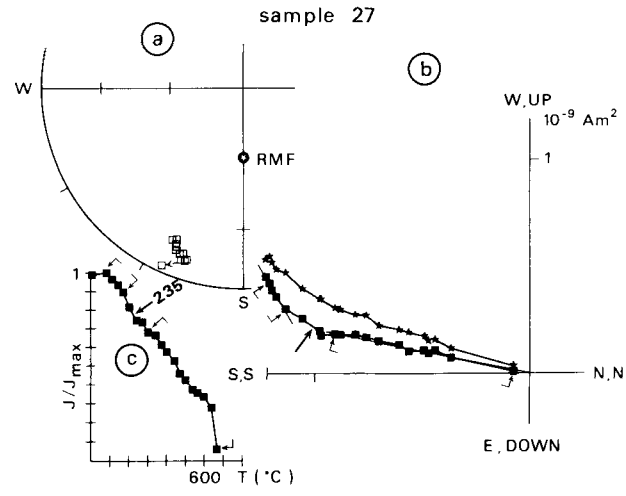


Figure 4. Same legend as for Fig. 2, for sample 27 (x = 93.5 cm from the dike border).

(ii) A Plio-Pleistocene overprint corresponding to intermediate T_{ub} whose spectral width varies greatly from one sample to the next, in opposite proportion to that of the Saxonian component.

(iii) A Saxonian primary magnetization carried by the grains with the highest T_{ub}.

In a few cases, the demagnetization was too noisy to allow us to determine the directions of the component.

The mean Plio-Pleistocene direction calculated on 22 sandstone samples falls 10.8° southwest from the direction recorded by the dike: $D = 217.7°$, $I = -48.0°$ (Table 1).

The averaged Saxonian direction is $D = 203.9°$, $I = -20.4°$ (Table 1). These rocks have been magnetized during the long Kiaman period of reversed magnetic polarity. The corresponding north VGP: $\varphi = 144.2°$E, $\lambda = 50.9°$N; $dp = 2.1$; $dm = 4.0$, falls within less than 9° of those obtained by Kruseman (1962), Evans & Maillol (1986) and Merabet & Guillaume (1988) on other upper Permian strata from the same area. It compares fairly well with the Eurasian APW path of the same age range (Irving & Irving 1982; Westphal *et al.* 1986).

The orthogonal projection diagrams can be used to determine two temperatures which delimit the transition between the Plio-Pleistocene and the Saxonian directions (Figs 2b, 3b and 4b):

(i) the maximum temperature at which only the Plio-Pleistocene overprint is removed; and

(ii) the minimum temperature above which only the Saxonian component remains.

Presupposing that the overprinted magnetization is not a CRM, we determine an apparent heating temperature, T_{app}, for each sandstone sample, which corresponds to the middle of the previously defined temperature interval, which then serves as the error bar. The apparent palaeotemperatures are given in Table 2. If no spurious magnetization has been acquired during the thermal treatment, the following two situations can happen.

(i) The change from the Plio-Pleistocene to the Saxonian direction is abrupt so that the error bar associated with T_{app}

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Table 2. Apparent palaeotemperatures recorded in the sandstone.

Sample	d	Palaeo-T_{app} (°C)
6	1.3	570±30
7	4.0	605±30
8	4.9	620±20
9	7.1	620±20
10	10.6	605±35
11	14.6	620±20
12	16.8	570±30
13	19.4	535±35
14	22.1	505±35
15	29.1	470±100
16	30.9	490±85
17	33.1	505±35
19	39.2	470±70
20	49.4	450±50
21	50.7	420±50
22	54.2	420±80
23	68.8	335±40
24	69.2	385±85
25	77.2	290±50
26	82.5	235±65
27	93.5	235±65
28	97.0	255±85
29	104.1	240±100
30	108.5	235±100
31	116.4	270±100
32	117.3	220±80
33	119.5	140±100

d : distance, in cm, from the dike/sandstone contact.

reduces to the interval between two consecutive heating steps (30° or 40 °C). In this case, there is no overlap of the unblocking temperatures and the Plio-Quaternary component is a pure PTRM.

(ii) The change in direction is progressive and regular; this may be indicative of the superposition of two magnetization components (Daly 1981; McClelland Brown 1982): a Plio-quaternary CRM overlapping with the original Saxonian CRM. As the unblocking temperature specturm of a CRM always extends from its acquisition temperature to some higher temperature, such as the Curie temperature (Tan Ik Gie and Biquand, 1988), in this unfavourable situation, the apparent palaeotemperature would be overestimated by half the width of its error bar. This emphasizes the need for determining the nature of the overprinted magnetization.

In the present study the error bars associated with T_{app} (Table 2) probably overestimate the true T_{ub} overlap. First, the minimum error bar is limited by the temperature interval (30° or 40 °C) between two consecutive demagnetization steps. Second, the error bars integrate the scatter in the magnetization directions due to the spurious magnetization; thus, an error bar of 100°–200 °C does not necessarily mean the T_{ub} overlap over this entire temperature interval.

The basic question then is whether these T_{app} represent true palaeotemperatures; in other words, is the overprinted magnetization a (P)TRM or a CRM? Also, because of differences between the natural and laboratory cooling rates, how much would the apparent palaeotemperature overestimate the true one?

4 NATURE OF THE PLIO-PLEISTOCENE OVERPRINTED MAGNETIZATION AND PALAEOTEMPERATURE PROFILES

We assume that prior to the dike intrusion, the sandstone was magnetically homogeneous. The development of a CRM implies the crystallization of new magnetic minerals in a rock. If we can prove that the injection of the dike did not produce new magnetic minerals, we will have demonstrated that the overprint is not a CRM and is therefore a partial or a total TRM. Various magnetic parameters sensitive to the chemical composition of the magnetic minerals will be used to study this problem. The distance x will always be measured from the dike/sandstone interface.

4.1 NRM intensity

The NRM intensity (J_{NRM}) in the sandstone (Fig. 5), drops in the 15 first cm (sample 6 to about 11), decreases regularly until 120 cm (sample 12 to 34), and is then stable beyond this distance. The trend observed in the middle part of the profile is consistent with the hypothesis of a TRM whose intensity decreases away from the dike. The intensity $J_{NRM} = 3.02 \times 10^{-5}$ A m^2 kg^{-1} measured in sample 6, next to the dike, is about an order of magnitude higher than in the more remote samples and in other red sandstones from the 'Bassin de Lodève'. In agreement with the demagnetization curve of this sampe, this suggests that recrystallization of magnetite has occurred at the dike contact.

4.2 Viscosity coefficient

The viscosity coefficient is generally weak in the vicinity of the dike, fluctuating around 4 per cent, and increases regularly up to 15 per cent from about $x = 22$ cm (sample 14) outwards. The VRM intensity, J_{VRM}, acquired during the 15 day experiment drops until $x = 17$ cm, then remains constant (Fig. 5). As this VRM is carried by the grains whose relaxation times are less than 15 days, these data indicate that beyond 17 cm (sample 12) the intrusion of the dike has not produced changes in the magnetic characteristics of these grains, but they do not provide information about the population which carries the 'stable' part of the remanence.

4.3 Initial magnetic susceptibility

The initial volume susceptibility, K, measured before heating (Fig. 5) is high in the samples closest to the dike (1.22×10^{-4} SI), decreases down to a minimum value at $x = 17$ cm (sample 12), then increases slightly before it stabilizes, around a mean value of $6.18 \pm 0.43 \times 10^{-5}$ SI from about $x = 22$ cm (sample 14) to 120 cm (sample 33). Farther away, the susceptibility values are more dispersed even varying between two specimens from the same sample; this probably reflects variations on a small scale of the concentration of magnetic minerals. The decrease in K observed at the beginning of the profile is consistent with the idea of magnetite recrystallization. However, the depletion in K around sample 12 is puzzling and does not support the idea that there is a transition zone of recrystallized magnetite from the dike border to about 17 cm (sample 12). Since the susceptibility of magnetite is significantly stronger than that of haematite, the susceptibility should not be lower closer to the dike ($10.5 < x < 19.5$ cm; samples 10 to 13) than farther from it, where probably less recrystallization has occurred. As the susceptibility essentially reflects the chemical composition

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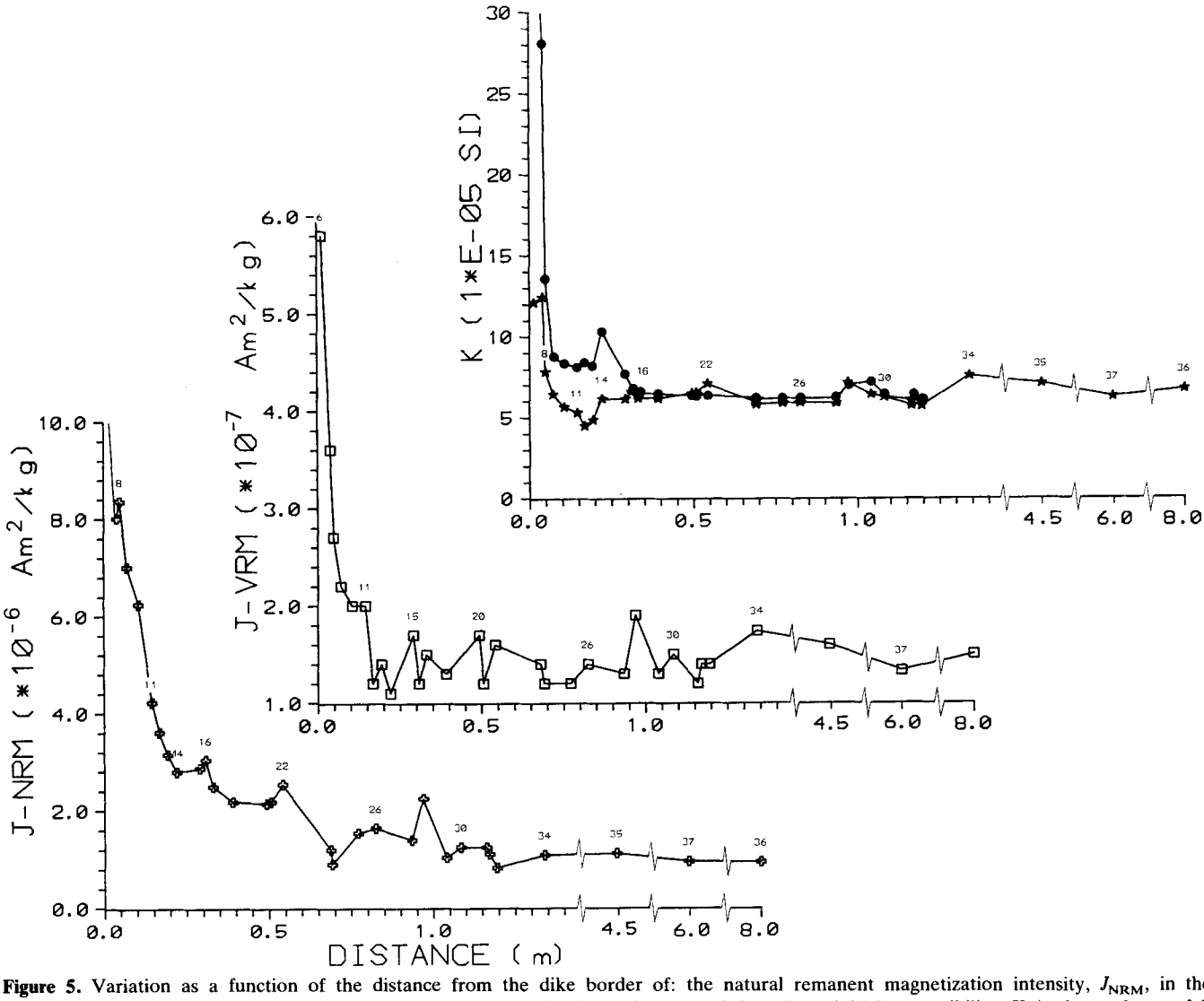


Figure 5. Variation as a function of the distance from the dike border of: the natural remanent magnetization intensity, J_{NRM}, in the sandstone; the viscous remanent magnetization intensity, J_{VRM}, in the sandstone; and the volume initial susceptibility, K, in the sandstone; full stars: before heating; full circles: after heating to 700 °C.

and the concentration of the magnetic minerals, we are led to the conclusion that recrystallization followed by a partial dissolution of the magnetic minerals might be a plausible explanation for the susceptibility low.

K was also measured after heating to 700 °C in order to check possible changes in the magnetic mineralogy resulting from the thermal treatment in air. The profile (Fig. 5) shows a strong increase in K for the two samples closest to the dike (samples 6 and 7), followed by a more regular decrease (except for the sample 14) as far as 29 cm (sample 15), beyond which the susceptibilities remain constant around values which are not significantly different from those measured before heating ($6.38 \pm 0.33 \times 10^{-5}$ SI for samples 16 to 33). The two susceptibility profiles strongly suggest that before about 29 cm (sample 15), recrystallization of magnetite occurred during thermal treatment, and that the sandstone was not chemically stabilized after the recrystallization and/or the dissolution generated by the dike injection. Conversion of haematite into magnetite upon heating in air has been observed in red sandstones above 620 °C by Dunlop (1972). Beyond $x = 29$ cm, the samples proved to be chemically stable when heated in air, as if they had not been affected by the dike intrusion.

4.4 Demagnetizing curves

Except for the first sample, all the demagnetization curves of the sandstone samples display an inflection point and a shape which is characteristic of the presence of two magnetic carriers such as magnetite and haematite (Figs 2c, 3c and 4c). The possibility that the magnetite was produced by the thermal treatment can be ruled out for the samples situated beyond 29 cm (sample 15) because their initial susceptibility is unchanged by the heating (Fig. 5). But was this magnetite originally present in the rock before the dike intrusion, or is it a secondary product of the magmatic event?

If we compare the demagnetization curves using the absolute values instead of the normalized ones, it is clear that the importance of magnetite decreases from the sample 7 until about sample 13 ($x = 19.5$ cm). Beyond this point,

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the curves become similar but still display the same pattern of a two-step decrease. This means that a small amount of magnetite was originally present in the sandstone. Magnetite also appears to be present in other red sandstones from the same area (Evans & Maillol 1986; Merabet & Guillaume 1988).

4.5 Hysteresis cycles

A common feature in all the sandstone samples is the shape of the induced magnetization curve recorded up to a maximum induction of 1.5 T (15 kG). All of the curves can be divided in two parts, although the first one is generally not obvious (Fig. 6): the first part seems to saturate in relatively low magnetic fields (0.3 to 0.4 T) and is indicative of a magnetic mineral with low coercivity like magnetite. In the second part, the coercivities are much higher. Because of the concave shape of the curves at the highest fields, it is clear that saturation is not being reached at 1.5 T. Saturation in fields higher than this value is common in samples containing natural haematite (Dunlop 1971, 1972; Dankers 1978; Dekkers 1988). We note that the induced magnetization curve of sample 6 (Fig. 6, curve 1) reveals the presence of haematite, which did not appear on its demagnetization curve.

The coercive force, H_c, and the remanent coercive force, H_{rc}, reflect the occurrence of haematite; H_c and H_{rc} are high (a few 10^5 A m^{-1} or several thousands Oe) which is consistent with a large population of highly coercive magnetic grains (haematite) and a minor amount of grains with lower coercivities (magnetite) (Day, Fuller & Schmidt 1977).

Because of the composite nature of the magnetic mineralogy, the hysteresis parameters can provide only qualitative information about, for example, their evolution as a function of the distance from the dike. The induced saturation magnetization, J_s, the remanent saturation magnetization, J_{sr}, H_{rc} and to a lesser degree H_c, change close to the dike until about 29 cm (sample 15) after which they remain roughly constant. Comparison of the first magnetization curves (Fig. 6) shows a more or less progressive evolution from the samples closest to the dike (curve 1), where the magnetization grows quickly and to high values, to the more remote samples, from $x = 0.30$ to 8 m (curve 3), where the increase in magnetization is slower and more limited. This is again consistent with the interpretation that the dike generated magnetite as far as 29 cm, but not beyond this distance.

Comparison of the induced magnetization curves obtained on specimens before heating and on other specimens from the same samples after heating to 700 °C, shows that in the samples nearest to the dike the saturation occurs earlier and at higher values for the heated specimen (Fig. 6, curve 1). Away from the dike, the hysteresis loops obtained before and after heating become closer and closer, and are almost identical in samples situated beyond about 29 cm. This result is in agreement with the susceptibility measurements and supports the idea that laboratory heating has produced some additional magnetite close to the dike but had no effect in the more remote sandstone samples which seem to have remained chemically stable since their Permian diagenesis.

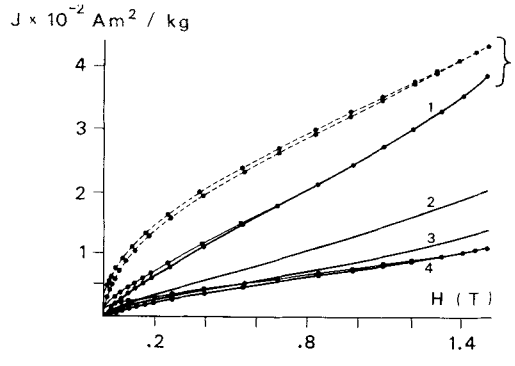


Figure 6. Induced magnetization curves versus the applied magnetic field; continuous lines: before heating; dashed lines: after heating to 700 °C. Half the hysteresis cycles of sample 6 (curves 1) and sample 12 (curves 4) are shown; for the sake of clarity, the curves recorded after heating are not drawn for this latter sample because they would almost merge with those obtained before heating. Schematically, the first magnetization curves correspond to four types: curve 1: samples situated at $1.3 \leq x \leq = 4.0$ cm from the dike border (samples 6 and 7); curve 2: $4.9 \leq x \leq 22.1$ cm: samples 8 to 14; and curve 3: $29.1 \leq x \leq 119.5$ cm: all the other samples from 15 to 36, except samples 12 ($x = 16.8$ cm) and 19 ($x = 39.2$ cm) represented by the curve 4.

4.6 Estimation of the relative proportion of haematite and magnetite

An estimate of the volumetric percentage of haematite and magnetite present beyond 29 cm can be made, using the averaged observed susceptibility. Taking 1.18×10^{-3} SI for the antiferromagnetic susceptibility of 100 per cent haematite (Néel & Pauthenet 1952; Craik 1975) and 2.5 SI as average initial susceptibility for 100 per cent magnetite (Stacey & Banerjee 1974), we calculate that 1 per cent of haematite mixed with only 0.0019 per cent of magnetite would account for the observed K value. Note that in the extreme hypothesis where haematite would be the only magnetic mineral present in the rock, a volumetric proportion of 5 per cent is required to give the measured susceptibility; such a high percentage is very unlikely, even in a red sandstone; hence the need of another magnetic mineral. The concentration of haematite and magnetite previously calculated is also consistent with the values of J_s calculated by extrapolation using a law in $1/H$.

4.7 Apparent palaeotemperature profile

Examination of all the observations from the dike/sandstone interface to 8 m distance leads to the following conclusions.

(i) Prior to the dike injection, haematite with a small amount of magnetite or Ti-poor titanomagnetite was present in the sandstone.

(ii) In the vicinity of the dike, as far as 29 cm (sample 15), the magmatic intrusion led to the crystallization of new magnetite, possibly followed by partial dissolution in a few samples. Consequently, these samples are likely to carry a CRM and are not suitable for palaeotemperature determinations. Simultaneously, the sandstone became chemically

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unstable, since additional magnetite can form during thermal treatment in air in the laboratory.

(iii) From about 30 cm (sample 16) to 8 m (sample 36), no noticeable recrystallization has occurred either by emplacement of the basaltic dike or by heating in the laboratory. Consequently, the Plio-Pleistocene overprinted magnetization of the samples can be assumed to be a partial TRM.

This conclusion apparently conflicts with the wide error bars associated with the apparent palaeotemperatures in the samples 16 to 33 ($30 < x < 120$ cm). Such error bars suggest large overlap of the Plio-Pleistocene and Saxonian T_{ub} spectra, which would usually be interpreted as evidence for Plio-Pleistocene CRM. As noted previously, noisly magnetization directions can account for part of this ovelap, but there is probably also some real overlap whose width is unknown. This overlap may not necessarily mean that a CRM is present. McClelland Brown (1982) underlines that T_{ub} overlap could also arise from the presence of PTRMs carried by multidomain and single-domain states of the same mineral, or from PTRMs carried by two coexisting magnetic minerals; in the present study, the first situation is possible while the second proved to be real. Also, it is reasonable to assume that the farther we go from the heat source, the fewer the metamorphic transformations, and consequently the less likely it is that CRM will occur. For example, Pullaiah *et al.* (1975) observed no chemical change in the magnetic mineral composition of rocks containing haematite when they were heated to 400 °C for 6 days. Thus, we accept the statement that the samples situated beyond $x = 30$ cm (sample 16) are suitable for palaeotemperature determinations.

Moreover, we can assume that in the area where the magnetite has recrystallized no haematite has been produced. This is a reasonable assumption because if the environmental conditions were reducing, oxidation reactions could not have occurred; in this case, the Plio-Pleistocene component of magnetization carried by these samples can only be a PTRM of haematite, as a CRM carried by magnetite would not have survived above 580 °C. Consequently all the samples whose apparent palaeotemperature is higher than the Curie point of magnetite ($T_{app} > 580$ °C) should be valid. This applies in particular to samples situated between $4 \leq x \leq 14.5$ cm (samples 7 to 11). Another argument is that the unblocking temperatures of a CRM carried by haematite would extend from the true palaeotemperature reached by the sample to almost the haematite Curie temperature; consequently, at the highest temperatures of the thermal treatment, one would expect to observe a magnetization direction intermediate between Plio-Pleistocene and Saxonian directions, instead of the unique Saxonian direction which is indeed recorded.

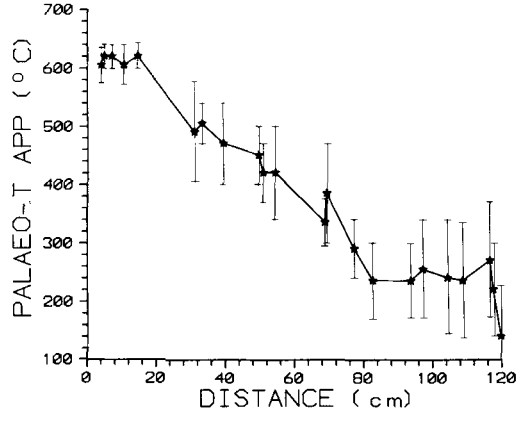


Figure 7. Apparent palaeotemperatures reached in the sandstone versus the distance from the dike border, and associated error bars (see text for explanations).

Thus, we arrive at the apparent palaeotemperature profile given in Fig. 7. The values plotted are those of Table 2 without samples 12 to 15 ($17 < x < 29$ cm). Sample 6 ($x = 1.4$ cm) has also been eliminated because it cannot give more than a minimum palaeotemperature estimate of 570 °C, since apparently no higher T_{ub}'s have been preserved.

4.8 Corrected palaeotemperature profiles

These apparent palaeotemperatures would be true palaeotemperatures if, in the field, the sandstone samples had been heated and subsequently cooled in conditions similar to those in the laboratory. But in nature, time constants are longer than experimental ones.

The principle of analogy between time and temperature follows from the Néel's theory for an assembly of single domain (SD) magnetic grains (Néel 1949; Dunlop & West 1969). A population of SD grains that has just been demagnetized at a temperature T_2 in a laboratory experiment over a short time t_2 will just be demagnetized, and possibly remagnetized, at a lower temperature T_1 acting in nature for a longer time t_1. This situation is expressed by Pullaiah *et al.* (1975), Enkin & Dunlop (1988) in the following equation:

$$T_2 \log Ct_2/F(T_2)G(T_2) = T_1 \log Ct_1/F(T_1)G(T_1),$$

where C is the characteristic frequency of thermal fluctuations ($=10^{10}$ Hz; Néel 1949) and $F(T)$ and $G(T)$ represent the temperature dependences of the saturation magnetization and the coercive force, respectively:

$$F(T) = J_s(T)/J_{s0}, \qquad G(T) = H_c(T)/H_{c0},$$

with the subscript 0 denoting room temperature value. The curves calculated by Pullaiah *et al.* (1975) using the previous equation allow us to determine the variation of the blocking temperature T_1 of the overprinted magnetization with the time t_1 during which T_1 is maintained on the field.

Using a different approach, Dodson & McClelland Brown (1980) and McClelland Brown (1982) calculate the blocking temperature dependence on the cooling rate.

To correct the apparent palaeotemperature values, we need to know, in the first case, how long the sandstone samples remained at their peak temperature $T_1(t_1)$, and in the second case how long it took to cool the samples (v_1). The answers are not simple since these parameters depend on the distance of the host rock from the heat source: the closer to the dike, the shorter the time t_1 and the faster the cooling after the magma has stopped flowing. Anticipating the results given by the thermal modelling used in this study, we calculate a maximum time t_1 which varies from 12 hr at $x = 5$ cm, to 8 days at $x = 120$ cm. We used these values to

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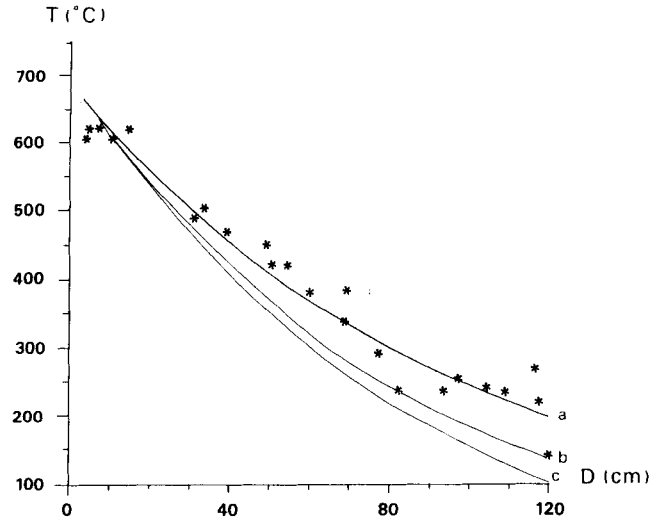


Figure 8. Curve (a): best fit curve of the apparent palaeotemperature profile through the experimental data (stars); curve (b): palaeotemperature profile corrected for the thermal fluctuation effect, according to Pullaiah *et al.* (1975); curve (c): palaeotemperature profile corrected for the cooling rate effect, according to Dodson & McClelland Brown (1980).

correct the apparent palaeotemperature profile according to Pullaiah *et al.* (1975) (Fig. 8, curve b).

Similarly, the maximum cooling rate, v_1, is found to vary from 7 °C day^{-1} at $x = 5$ cm, to 0.6 °C day^{-1} at $x = 120$ cm. These rates can be used to make corrections according to Dodson & McClelland Brown (1980) (Fig. 8, curve c). Longer time constants would accentuate the departure of the curves b and c from the apparent palaeotemperature profile, especially in the low-temperature ranges. Note that to draw these curves, we needed to make the simplifying assumption that haematite, in SD grains, was the only magnetic carrier in the sandstone.

5 THERMAL MODEL AND MAGMA FLOW DURATION

During a dike injection, the flowing magma advects heat through the dike, while heat is mainly conducted out of the dike into the surrounding rock. However, according to Delanay (1987), transient heat conduction can be applied to most dikes provided they are thin, despite the effects of magma flow during emplacement and groundwater circulation during cooling. Calculations of temperature in and around a magmatic intrusion are generally done with unidirectional unsteady heat conduction models, in which a stagnant sheet of magma at a uniform initial temperature is instantaneously brought into contact with a uniformly cool host rock (Lovering 1935; Carslaw & Jaeger 1959; Jaeger 1967). Other models allow the magma to flow a certain time, τ, before it solidifies (Carslaw & Jaeger 1959; Turcotte & Schubert 1982; Delanay & Pollard 1982; Delanay 1987; Fabre *et al.* 1989). However, these analytical models do not allow us to take into account various critical parameters which play an important role during the cooling process.

5.1 Thermal model

To account for the maximum parameters in the calculation of the temperature field in and around the dike, we have used a program based on the finite elements method which allows the effects of the following parameters to be simultaneously examined.

(i) The thermal diffusivity, $\lambda = k/\rho c$, of both the dike and the host rock, and its temperature dependence which enters through the temperature dependence of the thermal conductivity, $k = \alpha + \beta/T$ established from empirical considerations (Hanley, Dewitt & Roy 1978); the product of the density and the heat capacity of the rock, ρc, is assumed to be constant.

(ii) The latent heat of crystallization of the basalt, L_f.

(iii) The flow duration of the magma, τ.

(iv) The width of the flowing magma in the central part of the dike relative to the thickness of the possibly solidified basaltic layer at the channel periphery.

Moreover, this program allows us to consider various boundary conditions (prescribed flux, fixed temperatures or exchange with a fluid).

A complete description of the code which runs on microcomputers can be found in Bonneville (in preparation).

The physical model based on a 2-D box 100 m wide and 200 m high is described in Fig. 9. The abscissa $x = 0$ corresponds to the dike axis and the reference layer is placed at the level $y = -100$ m which corresponds to the probable maximum depth of burial for the considered sandstone bed at the time of intrusion. The medium is divided into 36 elements (two in the vertical plane and 18 in the horizontal one): eight belong to the basalt and 28 to the sandstone. The mesh is dense and the elements regularly spaced along x in the basalt and in the sandstone as far as $x = 1.60$ m; beyond that, the remaining elements are much

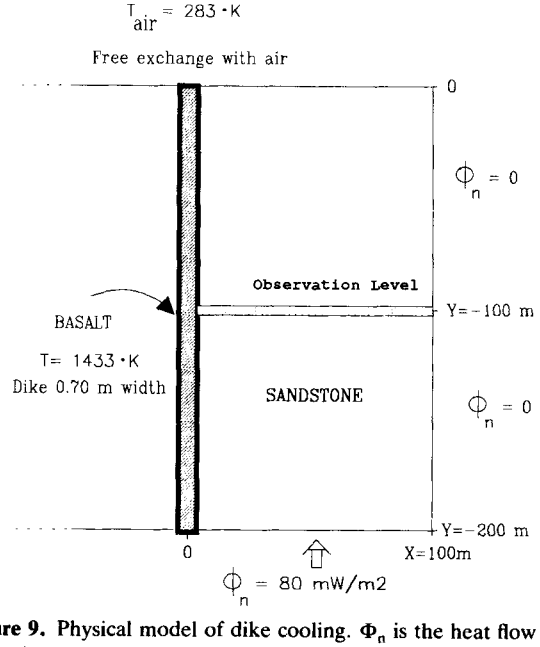


Figure 9. Physical model of dike cooling. Φ_n is the heat flow vector normal to the surface (see text for explanations).

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wider. Each element contains nine nodes, so that the temperatures be calculated over a total of 185 nodes, every 4.38 cm in the basalt and every 5 cm in the sandstone up to 1.60 m.

5.2 Boundary conditions and initial conditions

Before the dike intrusion, the flux is zero through the two vertical planes of the box ($x = 0$ and 100 m). At the bottom, we assume a vertical flux of 80 m W m^{-2} which corresponds to a normal geothermal gradient. At the top, we use a law of temperature exchange between the earth's surface and the air which gives a surface temperature fluctuating around 10 °C. Calculation of the temperature distribution in this static problem places the 12 °C isotherm at the sandstone level.

At $t = 0$ (iteration = 0), the dike is instantaneously emplaced and a temperature of 1160 °C (Fabre 1988) is imposed at all the nodes which now belong to the dike, while a contact temperature $T_c = 586$ °C (intermediate between 1160 °C and the host rock temperature), is assumed for the nodes belonging to the basalt/sandstone interface (Jaeger 1967; Delanay & Pollard 1982). From the first iteration onwards, T_c is no longer constrained to be 586 °C, and at short times the calculated contact temperatures converge toward the true one.

5.3 Experimental determination of some parameters

At room temperature, the thermal conductivity of a fresh basalt sample, k_{b0}, and of a sandstone sample from beyond the influence of the metamorphism, k_{s0}, are 1.80 W m^{-1} K^{-1} and 3.07 W m^{-1} K^{-1} respectively. The temperature dependences of k_b and k_s were then determined as $k_b = 0.02 + 522/T$ and $k_s = 0.57 + 732.9/T$, where T is in degrees Kelvin. The product, ρc, was obtained by measuring the densities and by estimating the heat capacities from Touloukian, Judd & Roy (1981): $\rho c_{basalt} = 2439500$ J m^{-3} K^{-1} and $\rho c_{sandstone} = 2358000$ J m^{-3} K^{-1}. The liquidus temperature, $T_l = 1160$ °C of the basalt was based on values obtained by Fabre (1988) for dikes outcropping in the same area. We assumed that the magma has a latent heat of crystallization of 9.0×10^8 J m^{-3}, released through a 210 °C temperature interval, from $T_l = 1160$ °C to $T_s = 950$ °C.

5.4 The different cases examined

The transient problem which results from the dike injection has been calculated for the following cases.

Case (1). The dike is injected instantaneously and cools.

Case (2). Subsequent to its injection, the dike is allowed to flow for a time τ, along its median plane $x = 0$, as in the model of Fabre *et al.* (1989). $T_l = 1160$ °C is thus maintained during τ at the 5 nodes whose abscissa is 0.

Cases (3) and (4). The dike is allowed to flow for τ through a certain width around its median plane ($T_l = 1160$ °C is maintained during τ at all the nodes belonging to this volume). These two cases representing magma flowing in a channel constricted by a solidified outer layer are probably the more realistic ones (Bruce & Huppert 1989). As mentioned in the field description, the dike is clearly divided in two parts. We assume that the central fresh zone, 45 to 50 cm wide, represents the channel of the flowing magma. Case (3) corresponds to a width of 52 cm, Case (4) to a width of 44 cm.

Each of the previous cases is subdivided into the following three subcases.

(a) The thermal conductivity is temperature independent ($k = k_0$) and there is no latent heat of crystallization ($L_f = 0$).

(b) The thermal conductivity is temperature dependent [$K = f(T)$] in the basalt and in the sandstone, and the latent heat of the basalt is included ($L_f > 0$).

(c) The effects of natural convection in the host rock is included the following way: when a fluid moves in a porous medium due to a temperature gradient, it carries some heat with an efficiency depending mainly on the gradient amplitude and on the permeability. The heat is transferred with an apparent conductivity all the higher since the convection is strong. Without any information about this convection, we have tried to simulate its consequences by majoring the lattice conductivity in the sandstone ($k = xk_0$, with x, the Nusselt Number, between 1 and 4) because natural convection can be approached by conduction mecanisms in which the apparent conductivity is majored. The Nusselt number always positive and defined as the ratio between this apparent conductivity and the lattice conductivity, quantifies the efficacy of this convection in conducting heat. The latent heat of the basalt is also considered.

A fourth situation (d) is also considered for Case (1) only: it is similar to Case (1a) except that the latent heat is included.

5.5 Results

The results are expressed as temperature versus time profiles for points situated in the sandstone at increasing distances from the dike border. The theoretical temperature profile is drawn using the maximum temperature reached at each point and is compared to the experimental and corrected palaeotemperature profiles of Fig. 8.

The palaeo-T curves (1a), (1b) (1c) and (1d) obtained for Case (1) (Fig. 10) are definitely too low compared to both the experimental palaeo-T_{app} data and the corrected palaeo-T curves.

Two remarks can be made concerning the temperatures reached by the sandstone close to the dike border:

(i) the effect of the latent heat alone is illustrated by the difference of about 100 °C between the curves (1a) ($L_f = 0$) and (1d) ($L_f > 0$); and

(ii) the temperature increase generated by the latent heat is roughly counterbalanced by the temperature decrease resulting from the thermal variation of the conductivities, as shown by curves (1a) ($L_f = 0$, $K = K_0$) and (1b) [$L_f > 0$, $K = f(T)$].

We also note that, except for Case (1d), our calculations give contact temperatures about 100 °C lower than the 586 °C imposed as an initial condition at the interface. This

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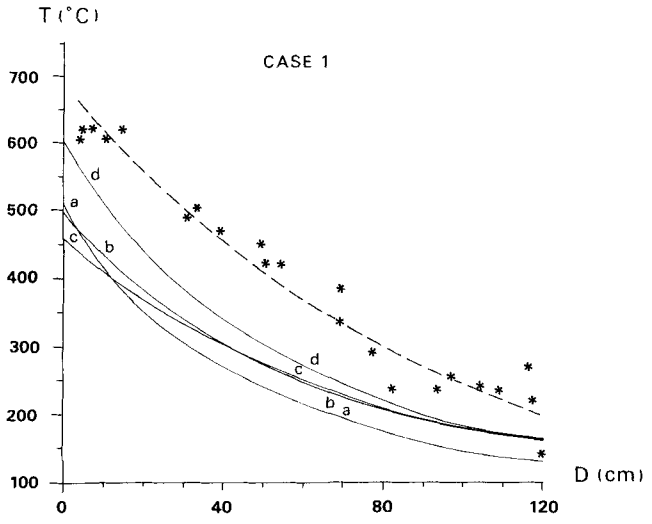


Figure 10. Temperature profiles calculated by the thermal model in Case 1 of no magma circulation ($\tau = 0$). Curve (a): $k_b = k_{b0} = 1.80\ W\ m^{-1}\ K^{-1}$; $k_s = k_{s0} = 3.07\ W\ m^{-1}\ K^{-1}$; $L_f = 0$. Curve (b): $k_b = f(T)$; $k_s = f(T)$; $L_f > 0$. Curve (c): $k_b = k_{b0}$; $k_s = xk_{s0} = 8.00\ W\ m^{-1}\ K^{-1}$; $L_f > 0$. Curve (d): same as curve (1a) but with latent heat.

is expected because the contact temperature is intermediate between the initial temperatures of the two bodies of rock, only if the thermal properties are identical on both sides of the interface. When they are different, as in this study, T_c can be calculated by the following equation (Jaeger 1967; Sacadura 1982, pp. 52–53), valid in the simplest case (1a), where thermal variation of the conductivity and latent heat are not taken into account:

$$T_c = (b_b T_b + b_s T_s)/(b_b + b_s),$$

where subscripts b and s refer respectively to basalt and sandstone, T is the initial temperature and $b = (k\rho c)^{1/2}$, with k = thermal conductivity, ρ = density, c = heat capacity. The analytical solution gives $T_c = 514\,°C$, which is in good agreement with the curve (1a) of Fig. 10 obtained by numerical analysis.

The curves calculated for Cases (1a, b, c, d) clearly show that a simple model involving instantaneous emplacement immediately followed by cooling is unrealistic for the present dike.

Case 2 is shown in Fig. 11. Although better than the previous case, the curve (2b) [$k = f(T), L_f > 0$ and $\tau = 10$ days] is still much too low in the proximity of the dike. Obviously, the result is worse when the flow duration is shorter. The fit is good in Case (a), for a flow duration of 3.25 days, and in Case (c) for $\tau = 3$ days, $L_f > 0$ and provided k_s does not exceed $k_{s0} = 3.07\ W\ m^{-1}\ K^{-1}$. We note that the curves calculated by our model compare well with the results obtained by the analytical solution proposed by Fabre *et al.* (1989) using the same values for the different parameters. In this model, the magma is also constrained to flow in the median plane of the dike, but the latent heat is neglected and the thermal diffusivity is assumed to be temperature independent and identical in the basalt and in the sandstone. The heat transfer in the host rock after the magma has stopped flowing is also taken into account. However, none of the curves of Fig. 11 (as well as others not shown here) provides a satisfactory fit to the two corrected profiles of Fig. 8.

The results for Cases (3) (Fig. 12) and (4) (Fig. 13) differ only slightly from one another. Cases (3a) and (4a) with no latent heat and $\tau = 1$ day do not fit the experimental data: the curves are too concave with temperatures much too high in the vicinity of the dike. The fit becomes better when the latent heat is included. The curves are still too concave for a width of 52 cm (curves 3b and 3c, Fig. 12) and the temperatures are too high near the interface, particularly when the conductivity is temperature dependent (curve 3b). Better fits are obtained for a 44 cm wide channel, for Case

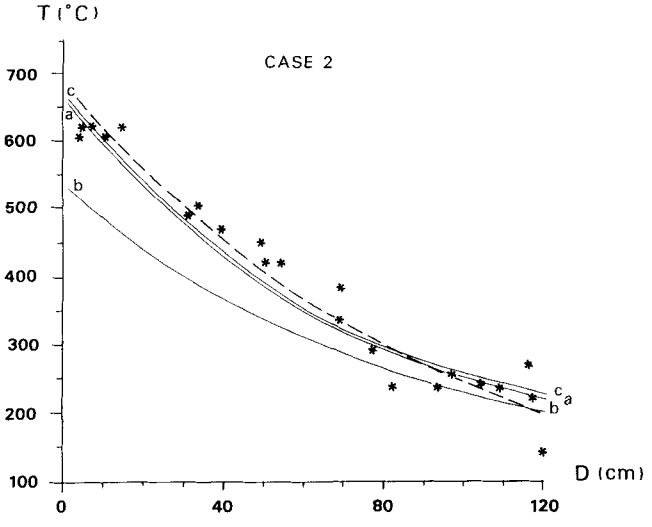


Figure 11. Temperature profiles calculated in Case 2 of a magma circulating along the median plane of the dike. Curve (a): $k_b = k_{b0}$; $k_s = k_{s0}$; $L_f = 0$; $\tau = 3.25$ days. Curve (b): $k_b = f(T)$; $k_s = f(T)$; $L_f > 0$; $\tau = 10$ days. Curve (c): $k_b = k_{b0}$; $k_s = 3.07\ W\ m^{-1}\ K^{-1}$; $L_f > 0$; $\tau = 3$ days.

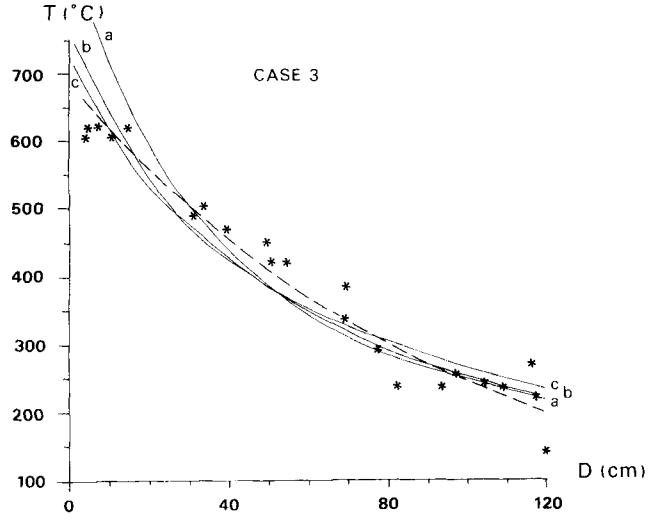


Figure 12. Temperature profiles calculated in Case 3 of a magma flowing through a 52 cm wide central channel. Curve (a): $k_b = k_{b0}$; $k_s = k_{s0}$; $L_f = 0$; $\tau = 1$ day. Curve (b): $k_b = f(T)$; $k_s = f(T)$; $L_f > 0$; $\tau = 3.25$ days. Curve (c): $k_b = k_{b0}$; $k_s = 6.0\ W\ m^{-1}\ K^{-1}$; $L_f > 0$; $\tau = 15$ hr.

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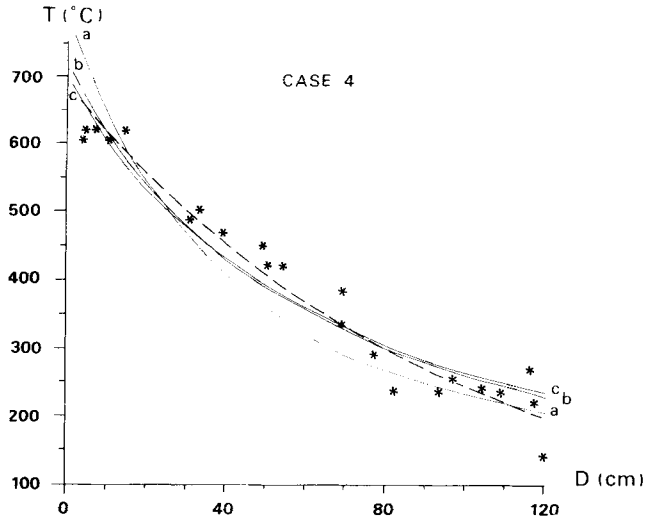


Figure 13. Temperature profiles calculated in Case 4 of a magma flowing through a 44 cm wide central channel. Curve (a): $k_b = k_{b0}$; $k_s = k_{s0}$; $L_f = 0$; $\tau = 1$ day. Curve (b): $k_b = f(T)$; $k_s = f(T)$; $L_f > 0$; $\tau = 5$ days. Curve (c): $k_b = k_{b0}$; $k_s = 5.36\ W\ m^{-1}\ K^{-1}$; $L_f > 0$; $\tau = 21$ hr.

(4b) which includes a thermal dependence for k [$k = f(T)$; $L_f > 0$ and $\tau = 5$ days] and for Case (4c) which includes limited natural convection in the sandstone ($k = 5.36\ W\ m^{-1}\ K^{-1}$; $L_f > 0$ and $\tau = 21$ hr).

We have also simulated the natural convection inside the flowing magma and have increased the basalt's thermal conductivity by a factor up to 4. As in Case 4(c), the width of injection is 44 cm, L_f is included, but $k_s = k_{s0} = 3.07\ W\ m^{-1}\ K^{-1}$. As expected, an increase of k_b produces an increase of heat accumulation in the sandstone next to the dike border, so that the temperatures are always much too high.

If convection is simulated simultaneously in the basalt and in the sandstone, the closer k_b to k_{b0}, the better the results, and this amounts to the previous Case (4c) with convection in the sandstone only.

5.6 Discussion

One of the aims of this study was to determine the flow duration of the dike. From the point of view of the discriminant role played by various parameters on the shape of the curves, the results appear a little disappointing; as a matter of fact the different cases examined do not lead to a unique solution: the thermal profiles are close to each other and reasonably fit the experimental data in the six different situations synthesized in Table 3. The following two basic points stand out from this table and from Figs 11, 12 and 13.

(i) The model's calculations generally fit the experimental data better when the latent heat is taken into account ($L_f > 0$).

(ii) If we eliminate Case 2 because it seems physically unrealistic to have magma flowing strictly along the median plane of a dike and because it does not agree with the field observations, we find two possible solutions: the first one includes thermal variation of k (Case 4b) and involves a flow duration of 5 days. The second one simulates a limited convection process in the sandstone only (Case 3c, or better 4c), and corresponds to significantly shorter flowing times of 15 and 21 hr respectively.

The two possible solutions refer to the experimental data without any correction. Whatever the values used for the different parameters, none of the curves calculated by the model fits the corrected palaeo-T profiles of Fig. 8.

Other temperature profiles can be obtained by changing the boundary conditions. For example, we can increase the temperatures by assuming a locally raised geothermal gradient due to the magmatic activity itself. A heat flux of $2.64\ W\ m^{-2}$ ensures a temperature of 100 °C at the sandstone depth. The resulting curves, calculated for Cases (4b) and (4c), are displaced toward higher temperatures only slightly close to the dike, but more and more at greater distances. The fit with the experimental data becomes poor, while the departure from the corrected profiles is even more pronounced than before. The hypothesis of an originally hot country rock, as McClelland Brown (1981) deduced from her study, is clearly unrealistic here and can be discarded.

The discrepancy between the curves calculated by the thermal model and the corrected palaeo-T curves still remains unsolved. There is no doubt that the differences between time constants used in the laboratory and those acting in nature must be taken into account. Whatever the method used, based on Néel's theory for SD grains, the corrected palaeotemperatures should indeed fall below the apparent palaeotemperatures, with a trend similar to that of the curves b (Pullaiah *et al.* 1975) or c (Dodson & McClelland Brown 1980) of Fig. 8. If the relaxation time versus blocking temperature curves of Middleton & Schmidt (1982) were used to correct the palaeo-T_{app}, the departure of the corrected palaeo-T from the former ones would even be more pronounced than for curve c (Fig. 8). Various tests

Table 3. Interesting studied cases.

	L_f	k_b	k_s	w	τ
Case 2a	No	1.80	3.07	x=0	3.25 days
Case 2c	Yes	1.80	3.07	x=0	3 days
(Case 3b	Yes	f(T)	f(T)	52	3.25 days)
Case 3c	Yes	1.80	6.00	52	15 hours
Case 4b	Yes	f(T)	f(T)	44	5 days
Case 4c	Yes	1.80	5.36	44	21 hours

This is a summary of the different cases examined and fitting more or less (line within brackets) the experimental palaeo-T_{app} data. L_f: latent heat of crystallization of the basalt; k_b and k_s are respectively the basalt's and the sandstone's thermal conductivity (in $W.m^{-1}.K^{-1}$); w: flowing width (in cm); τ: magma flow duration.

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carried out by Kent & Miller (1987) on redbeds support the model of Pullaiah *et al.* (1975). The discrepancy between the curves calculated by the thermal model and the corrected palaeo-T curves could be partly due to the fact that the corrections were made assuming that haematite was the only magnetic carrier, whereas some magnetite is probably also present. Moreover, the curves are established for strictly SD magnetic grains and if even a small amount of MD grains were present in the sandstone, they could significantly alter the relationship between T_{ub} and T_b. But the main reason could be that we overcorrected the apparent palaeotemperatures by using the longest time constants supplied by the model calculations. The times spent at each peak temperature and the cooling rates were based on Cases b $[k = f(T)]$.

The duration τ, required in Cases (b), would probably be increased if the model allowed the total variation of the diffusivity, λ, to be included, for example by adding the thermal variation of the heat capacity, (c). As a matter of fact, figs 12-101 and 12-102 in Touloukian *et al.* (1981) show that, with increasing temperature, the diffusivity of a sandstone like the present one, decreases twice as fast as the conductivity does.

These arguments lead us to favour Case (c), where a higher conductivity of the sandstone leads to a more rapid heat transfer and consequently a faster cooling, while the shorter flow duration ensures a shorter period at the peak temperature. If we use a flow duration of 15 hr (Case 3c) or 21 hr (Case 4c), the palaeo-T profiles corrected according to Pullaiah *et al.* (1975) and Dodson & McClelland Brown (1980) are closer to the palaeo-T_{app} data and consequently are in closer agreement with the calculated curves.

Finally, the tests concerning the effects of magmatic convection clearly indicate that this process do not play a crucial role in our problem.

From the above discussion and from examination of Figs 12 and 13, we conclude that the more likely conditions of this dike emplacement were a flow duration of approximately 21 hr, through a channel about 44 cm wide, with a thermal conductivity of $5.36\,W\,m^{-1}\,K^{-1}$ $(1.7\,k_{s0})$ in the sandstone and $1.80\,W\,m^{-1}\,K^{-1}$ (k_{b0}) in the basalt.

6 CONCLUSION

The thermal modelling clearly demonstrates that an instantaneous injection without magma flow is unrealistic for the present dike. This is in agreement with the lack of a glassy margin at the outer dike surface which, according to Fabre *et al.* (1989), indicates that the initially quenched glass has been destroyed by devitrification or by softening. Destruction of the glass can occur when the temperature of the dike margin reaches or exceeds the deformation temperature of the glass, determined to be $705° \pm 5\,°C$ by Lebeau & Girod (1988). In fact, Cases (3b), (3c), (4b), and (4c), but not (2a) nor (2c), all involve a temperature higher than 700 °C at the dike border. This result is also consistent with the process of melting back of a chilled margin described by Bruce & Huppert (1989).

It is satisfying that the best fits of the thermal curves are obtained in the more realistic situation of a magma flow of finite duration, with a width of flowing magma consistent with field observations, and with the latent heat of crystallization of the basalt taken into account. However, assuming a 44 cm wide channel, two different solutions can account for the experimental palaeo-T_{app} data.

In the first, we assume a thermal variation of the conductivity in both the basalt and the sandstone, and the flowing time, τ, required is 5 days; but in this case, the temperatures calculated by the thermal modelling never manage to fit the corrected palaeo-T curves.

In the second solution, we assume that heat is partly transferred by convection in the sandstone, and we simulate the circulation of pre-existing fluids by a moderate increase of the conductivity ($k_s = xk_{s0}$, with $1.5 \leq x \leq 1.7$); here a flow duration of only 21 hr is needed, which leads to a significantly better agreement with the corrected palaeo-T curves.

This study emphazises the necessity to take into account the magma flow duration in the thermal modelling; it also suggests that the thermal effects of a dike being intruded could simply be described by conductive heat transfer in the magma and by a limited natural convection in the host rock, as for example, the circulation of a pre-existing ground water.

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